Question 01

t	$v(t)$
2	10
4	20
6	25

(a) Degree, $n = 3-1 = 2$

$$P_2(n) = a_0 t^0 + a_1 t^1 + a_2 t^2$$

$$2^0 a_0 + 2^1 a_1 + 2^2 a_2 = 10 \quad \Rightarrow 1a_0 + 2a_1 + 4a_2 = 10$$

$$4^0 a_0 + 4^1 a_1 + 4^2 a_2 = 20 \quad \Rightarrow 1a_0 + 4a_1 + 16a_2 = 20$$

$$6^0 a_0 + 6^1 a_1 + 6^2 a_2 = 25 \quad \Rightarrow 1a_0 + 6a_1 + 36a_2 = 25$$

$$\begin{vmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{vmatrix} \begin{vmatrix} a_0 \\ a_1 \\ a_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 20 \\ 25 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a_0 \\ a_1 \\ a_2 \end{vmatrix} = \begin{vmatrix} -5 \\ 35/4 \\ -5/8 \end{vmatrix}$$

$$P_2(t) = -5 + \frac{35}{4}t - \frac{5}{8}t^2$$

$$P'(t) = \frac{35}{4} - \frac{5}{4}t$$

$$P''(t) = -\frac{5}{4}$$

$\therefore$ Acceleration, $P''(8) = -\frac{5}{4} = -1.25\ ms^{-2}$

(b)

$$P_2(t) = l_0(t) f(t_0) + l_1(t) f(t_1) + l_2(t) f(t_2)$$

$$l_0(t) = \frac{(t-t_1)(t-t_2)}{(t_0-t_1)(t_0-t_2)} = \frac{(t-4)(t-6)}{(2-4)(2-6)} = \frac{(t-4)(t-6)}{8}$$

$$l_1(t) = \frac{(t-t_0)(t-t_2)}{(t_1-t_0)(t_1-t_2)} = \frac{(t-2)(t-6)}{(4-2)(4-6)} = \frac{(t-2)(t-6)}{-4}$$

$$l_2(t) = \frac{(t-t_0)(t-t_1)}{(t_2-t_0)(t_2-t_1)} = \frac{(t-2)(t-4)}{(6-2)(6-4)} = \frac{(t-2)(t-4)}{8}$$

$$P_2(t) = \frac{(t-4)(t-6)}{8} \times 10 + \frac{(t-2)(t-6)}{-4} \times 20 + \frac{(t-2)(t-4)}{8} \times 25$$

(c) If a new data point is added in the above scenario, we need to use Newton's divided difference method. Because we don't have to calculate everything again in this method, just calculate the things for newly added data point.

The degree of the new polynomial $= 4-1 = 3$

Question 02

$f(x) = x\sin x$

ⓐ

	x	$f(x)$
x_0	$-\pi/2$	$-\frac{\pi}{2}\sin\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$
x_1	0	$0.\sin(0) = 0$
x_2	$\frac{\pi}{2}$	$\frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$

Degree, $n = 3-1 = 2$

$$P_2(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$$

$$= \frac{\pi}{2} - 1\left(x+\frac{\pi}{2}\right) + \frac{2}{\pi}\left(x+\frac{\pi}{2}\right)x$$

$$= \frac{\pi}{2} - x - \frac{\pi}{2} + \left(\frac{2x}{\pi} + 1\right)x$$

$$= -x + \frac{2x^2}{\pi} + x$$

$$= \frac{2}{\pi}x^2$$

$x_0 = -\frac{\pi}{2} \quad f[x_0] = \frac{\pi}{2}$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0 - \pi/2}{0 + \pi/2} = -1$$

$x_1 = 0 \quad f[x_1] = 0$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{\pi/2 - 0}{\pi/2 - 0} = 1$$

$x_2 = \frac{\pi}{2} \quad f[x_2] = \frac{\pi}{2}$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = \frac{0 - \pi/2}{\pi - \frac{\pi}{2}} = -1$$

$x_3 = \pi \quad f[x_3] = 0$

$f[x_0, x_1, x_2]$

$f[x_1, x_2, x_3]$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{1+1}{\pi/2 + \pi/2} = \frac{2}{\pi}$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{-1-1}{\pi - 0} = \frac{-2}{\pi}$$

$f[x_0, x_1, x_2, x_3]$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-\frac{2}{\pi} - \frac{2}{\pi}}{\pi + \frac{\pi}{2}} = -\frac{8}{3\pi^2}$$

(b) $x = \frac{\pi}{3}$

$$P_2\left(\frac{\pi}{3}\right) = \frac{2}{\pi}\left(\frac{\pi}{3}\right)^2$$

$$= \frac{2}{\pi} \cdot \frac{\pi^2}{9}$$

$$= \frac{2\pi}{9}$$

(c) new node $= \pi$

$f(\pi) = \pi \sin \pi = 0$

$n = 4 - 1 = 3$

$$P_3(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$
$$+ f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2)$$

$$= \frac{2}{\pi}x^2 - \frac{8}{3\pi^2}\left(x + \frac{\pi}{2}\right)(x + 0)\left(x - \frac{\pi}{2}\right)$$

$$= \frac{2}{\pi}x^2 - \frac{8x}{3\pi^2}\left(x^2 - \frac{\pi^2}{4}\right)$$

$$= \frac{2x^2}{\pi} - \frac{8x^3}{3\pi^2} + \frac{8x\pi^2}{4.3\pi^2}$$

$$= \frac{2x^2}{\pi} - \frac{8x^3}{3\pi^2} + \frac{2x}{3}$$

Question 03

$$f(x) = e^{3x} - e^{-3x}$$

$$\xi \in [-1.5, 1.5]$$

$$x_0 = 1, \quad x_1 = 0, \quad x_2 = 1$$

$$f(x) = e^{3x} - e^{-3x}$$

$$f'(x) = 3e^{3x} + 3e^{-3x}$$

$$f''(x) = 9e^{3x} - 9e^{-3x}$$

$$f'''(x) = 27e^{3x} + 27e^{-3x}$$

$$= 27(e^{3x} + e^{-3x})$$

$$f'''(1.5) = 27(e^{3\times1.5} + e^{-3\times1.5})$$

$$= 2430.762488$$

$$w(x) = (x - x_0)(x - x_1)(x - x_2)$$

$$= (x-1)(x-0)(x-1)$$

$$= (x^2 - x)(x-1)$$

$$= x^3 - x^2 - x^2 + x$$

$$= x^3 - 2x^2 + x$$

$$w'(x) = 3x^2 - 4x + 1$$

$$\Rightarrow 3x^2 - 4x + 1 = 0$$

$$\therefore x_1 = 1, \; x_2 = \frac{1}{3}$$

x	$w(x)$
1	0
$\frac{1}{3}$	0.14815
-1.5	-9.375
1.5	0.375

$$\text{error bound} = \left| \frac{f^3(\xi)}{3!} \right| \left| (x-x_0)(x-x_1)(x-x_2) \right|$$

$$= \left| \frac{2430.762488}{3!} \right| \times \left| -9.375 \right|$$

$$= 405.1270813 \times 9.375$$

$$= 3798.066388$$

$$= 3798.0$$